

Exercise task 1 (Basic): Subword tokenization

Part 1:

Text I chose for this exercise:

En: "According to all known laws of aviation, there is no way a bee should be able to fly. Its wings are too small to get its fat little body off the ground. The bee, of course, flies anyway because bees don't care what humans think is impossible. Yellow, black. Yellow, black. Yellow, black. Yellow, black. Ooh, black and yellow! Let's shake it up a little. Barry! Breakfast is ready! Coming! Hang on a second. Hello?"

Fin: "Kaikkien tunnettujen ilmailun lakien mukaan mehiläisen ei pitäisi mitenkään pystyä lentämään. Sen siivet ovat liian pienet nostamaan sen pientä, pulleaa ruumista ilmaan. Mehiläinen tietysti lentää siitä huolimatta, sillä mehiläiset eivät välitä siitä, mitä ihmiset pitävät mahdottomana. Keltainen, musta. Keltainen, musta. Keltainen, musta. Keltainen, musta. Oi, musta ja keltainen! Ravistellaanpa vähän. Barry! Aamiainen on valmis! Tulen! Odota hetki. Haloo?"

Translated with DeepL.com (free version)

English text had 98 tokens, 77 words, and 413 characters. Finnish text had 145 tokens, 58 words, and 459 characters.

Ratio: 58:77

What I noticed from the text and token amounts, is that in the english text there are less tokens but more words. However, in finnish, there are more tokens but not that as much words. This is because, finnish language relies on word conjugations (subwords), that are added to the amount of tokens. And this can lead to higher need for tokens and pricing.

Part 2:

```
In [1]: # Importing dependencies
import wordfreq as wf
from transformers import AutoTokenizer
from collections import Counter
```

```
In [2]: # getting the tokenizer
tokenizer = AutoTokenizer.from_pretrained("distilbert/distilbert-base-uncased")
```

```
In [3]: # getting the wordlist
wordlist: list[str] = wf.top_n_list(lang="en", n=100000)
```

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In [4]: # making the list to string for tokenizer
wordlist_as_string: str = ' '.join(wordlist)
```

```
In [5]: tokenized_list: list[str] = tokenizer.tokenize(wordlist_as_string)
```

Token indices sequence length is longer than the specified maximum sequence length for this model (215663 > 512). Running this sequence through the model will result in indexing errors

```
In [6]: print(len(tokenized_list))
```

215663

```
In [7]: for i in range(20):
        print(tokenized_list[i])
```

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and
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this
was
be
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have

```
In [8]: # top tokens
token_counter = Counter(tokenized_list)

print("Number of tokens in total:", token_counter.total())
print("Most common tokens:")
for item in token_counter.most_common(20):
    print(item)
print("Vocabulary size:", len(token_counter))
```

Number of tokens in total: 215663

Most common tokens:

('##s', 6182)
('s', 4136)
("'", 4129)
('##ing', 1495)
('##e', 1304)
('##a', 1268)
('##y', 1082)
('##d', 1077)
('##ed', 1068)
('##er', 980)
('##t', 976)
('##r', 964)
('##o', 901)
('##n', 879)
('##ly', 866)
('##i', 799)
('##l', 708)
('un', 642)
('##es', 576)
('.', 543)

Vocabulary size: 26536